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Scintillation studies of $Bi_4Ge_3O_{12}$ (BGO) down to a temperature of 6 K

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ABSTRACT

The scintillation light response and the decay time constant of $Bi_4Ge_3O_{12}$ (BGO) were measured down to a temperature of 6 K using a multi-photon counting technique. The light response of BGO was confirmed to be temperature independent below 100 K and an estimate of the light yield was made, giving a value of 23,700 ± 2600 ph/MeV. The temperature dependence of the scintillation decay time constant is interpreted using a simple three-level model for the emission centre which agrees well with the experimental data over the temperature range investigated.

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1. Introduction

Despite of rather low light yield at room temperature (not more than 20% of NaI-Tl [1,2]) bismuth germanate (BGO) is regarded as a very important scintillator material, due to its very good radiation length ($X_o = 1.12$ cm), reasonably short decay time (a fraction of a microsecond), low afterglow, mechanical stability and an emission spectrum matching well that of semiconductor photodetectors. Soon after its characterisation as a scintillator in the early 1970s [3,4] it became the material of choice of high resolution Positron Emission Tomography (PET) [5]. In this application the light yield is obviously a limiting factor and efforts were made to increase this parameter.

BGO is an intrinsic scintillator and its emission is strongly affected by a thermal quenching process; a decrease of temperature leads to an increase of luminescence intensity. Therefore, cooling was identified as one of the ways to improve the performance of the crystal as a scintillator. The change of BGO scintillation properties with temperature has been studied by Piltingsrud down to 77 K using a photomultiplier tube (PMT) and it was shown that a threefold increase of the BGO light yield over that at room temperature can be achieved [6]. This finding was recently confirmed [2] using an avalanche photodiode (APD) as light detector. The lack of reliable photodetector that can operate over a wide temperature range precluded studies of scintillation properties below 77 K and no further low-temperature work has been reported, except merely qualitative observations of BGO scintillations at liquid helium temperature, using solid-state photon counters [7].

The interest in low-temperature studies of scintillation materials has received a new impetus recently with the development of hybrid cryogenic phonon-scintillation detectors (CPSD), operating at millikelvin temperatures. These detectors combine a very low energy threshold and an excellent resolution with the ability to distinguish different types of particle interactions [8]. CPSD have proven their advantage in experimental search for extremely rare events, such as the interaction of Weakly Interacting Massive Particle (WIMP) dark matter with nuclei, neutrinoless double-beta decay or radioactive decay of long-lived nuclei (see Ref. [9] and references therein). The power of the technique can be appreciated by looking at the value for the half-life of natural ^{209}Bi ($1.9 \pm 0.2 \times 10^{19}$ yr) that has been measured recently by some of the authors using CPSD with a BGO target [10].

The prospect of application of the CPSD technique in the multi-target cryogenic search for WIMP dark matter, EURECA[1] [11], has intensified research aiming at characterising scintillation properties of different materials down to cryogenic temperatures [9]. In this respect, bismuth germanate is thought to be a very important target, permitting a cross comparison with other detectors that use germanium semiconductors such as cryogenic phonon detector [12] or operating in conventional regime [13]. Due to the above and due to a lack of knowledge of the low-temperature scintillation properties, we initiated studies of scintillation light response and decay time constant of BGO as a function of temperature down to 6 K.

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[1] European Underground Rare Event Calorimeter Array, www.eureca.ox.ac.uk.

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2. Experiment

The measurements of scintillation characteristics of BGO were carried out using the multi-photon counting (MPC) technique. A detailed description of the basic experimental set-up and the data analysis procedure is given in the original paper [14] and subsequent papers which present progressive improvement of the technique [15,16].

For this study we used a $5 \times 5 \times 1\,mm^3$ BGO crystal purchased from Amcrys-H (Kharkiv, Ukraine). The crystal was mounted to the holder of a conventional He-flow cryostat (Oxford Instrument). An ^{241}Am source (33 kBq) is located at ~2 mm distance from the sample. To reduce pile-up, we used a stainless steel aperture with a hole of 0.1 mm diameter between the source and the sample. The photon pulses produced by scintillation processes were detected by a green-sensitive PMT (model 9124A, Electron Tubes), recorded and analysed offline. The light yield is deduced from a histogram of the scintillation pulse height and the decay characteristics are obtained from the distribution of arrival times of individual photons.

3. Results and discussion

Fig. 1 shows the temperature dependence of the scintillation light response of BGO between 6 and 307 K for α-particle excitation. As temperature decreases to 100 K the scintillation light output increases steadily. At lower temperature, however, the scintillation response remains virtually constant. The quenching temperature T_q (temperature, at which the light output exhibits half of its maximal value) is found to be 250 K and the total light output of the scintillator increases by a factor of ~3.5 from room temperature ($T = 295$ K) to 6 K. This behaviour is consistent with what was reported in earlier studies of the luminescence properties of the crystal using optical excitations [3,17].

Provided that the ratio of scintillation signal amplitudes at a given excitation energy of the α-particles and γ-quanta (α/γ ratio) for BGO shows no noticeable temperature dependence, the measured temperature dependence of the light response of BGO at α-particle excitation can give a good estimate for the light yield of the crystal in the low-temperature range.

Earlier studies by Sysoeva et al. [18] demonstrated that this assumption is valid for BGO over the 253–350 K temperature range, where the α/γ ratio remains constant (~0.20). Recent cryogenic experiments with a BGO target [10], carried out at 20 mK gave an α/γ ratio equal to 0.21 at 5.4 MeV and 0.17 at 3.2 MeV. The first value measured at the same energy (5.4 MeV) agrees well with the data of Ref. [18] whereas the reduction of the α/γ ratio with lower α-particle energy is due to the energy dependence of the α-particle's ionisation density [19], being proportional to the length of ionisation track [20]. The same α/γ ratio ~0.20 was found for $CaWO_4$ at room temperature [21] and at few mK [22]. The results of studies of $CdWO_4$ are also consistent with this interpretation; the α/γ ratio is reported to be 0.18 at room temperature [19] and also at 12 mK [23]. Admittedly the temperature independence of the α/γ ratio is a rather common feature of intrinsic oxide scintillators. Assuming therefore the α/γ ratio of BGO as temperature independent and taking the light yield of BGO at room temperature to be 6900 ± 140 ph/MeV [2] one can estimate the light yield of the crystal at 6 K as being $23{,}700 \pm 2600$ ph/MeV. Given that the light response of the BGO does not change below 100 K, this number is very consistent with a value of $22{,}700 \pm 2000$ ph/MeV at $T = 77$ K, quoted by Moszynski et al. [2]. The observed temperature dependence of the BGO light yield is a common feature of intrinsic scintillators [9]; it manifests a classic effect of the thermal quenching governed by thermally activated non-radiative processes that compete with the radiative ones.

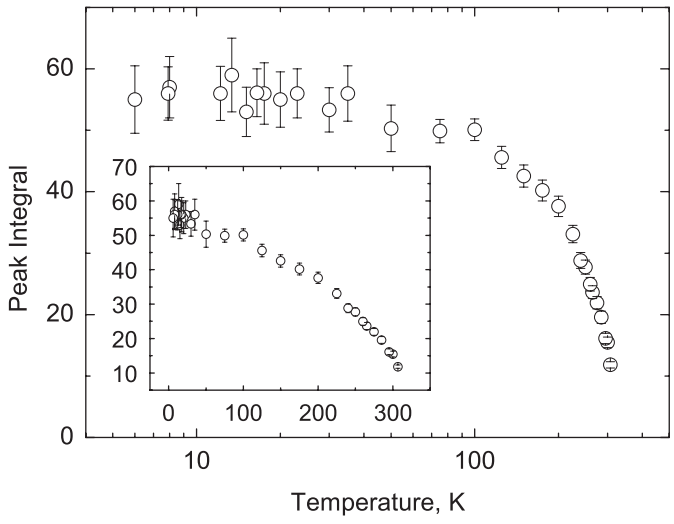


Fig. 1. Temperature dependence of the scintillation light response of BGO under α-excitation (^{241}Am). The inset shows the dependence on a linear temperature scale.

Since the luminescence decay is controlled by the same processes, the crystal exhibits a strong temperature dependence of this characteristic. To the best of our knowledge, the low-temperature scintillation decay constant of BGO has been reported only for 195 [6] and 4 K [7]. The use of the MPC technique allowed to carry out the measurements of this characteristic over the entire temperature range of interest. Fig. 2(a–c) shows the scintillation decay curves of BGO measured at 295, 77 and 6 K under α excitation, respectively. The decay time at 295 K was fitted by a single exponential function and τ was found to be 0.43 ± 0.08 μs, while the room-temperature scintillation decay time constant of BGO is 0.3 μs [4–6]. It should be mentioned that while being well suited for measurements of slow scintillation processes and temperature dependences thereof, the MPC technique becomes less precise when the main decay time constant decreases to values below 1 μs [14]. This limitation arises from the time resolution of the experimental set-up currently in use.

Fig. 3 shows the variation of the decay time constants with the temperature. At first, the cooling of the sample from 300 to 150 K causes a steep increase of the decay time constant. This is the region where the kinetics of the radiative decay is strongly affected by thermal quenching and the decay time curve is nearly a single exponential function. Below 150 K the non-radiative processes become less efficient and therefore the augmentation of the time constant is less pronounced. At this temperature, the decay splits into two components and the best fit of the experimental data was obtained using a sum of two exponential functions. The results of fitting for 77 and 6 K are listed in Table 1. The character of the $\tau = f(T)$ dependence changes again abruptly below 50 K: the decay kinetics exhibits steep retardation until the temperature reaches 10 K. The character of the temperature dependence of the scintillation decay is very similar to that measured by Moncorge et al. [17] and Rogemond et al. [24], using ultraviolet excitation. It is also worthwhile remarking that the measured decay time constant at $T = 200$ K is 2.2 ± 0.2 μs, consistent with the number given by Piltingsrud [6] (2 μs at 195 K). Furthermore, the value of $\tau = 138$ μs at $T = 6$ K obtained by us is very close to the estimate made by Cho et al. [7] (150 μs).

The temperature dependence of luminescence properties of BGO has been the subject of extensive studies [3,17,24–28] and

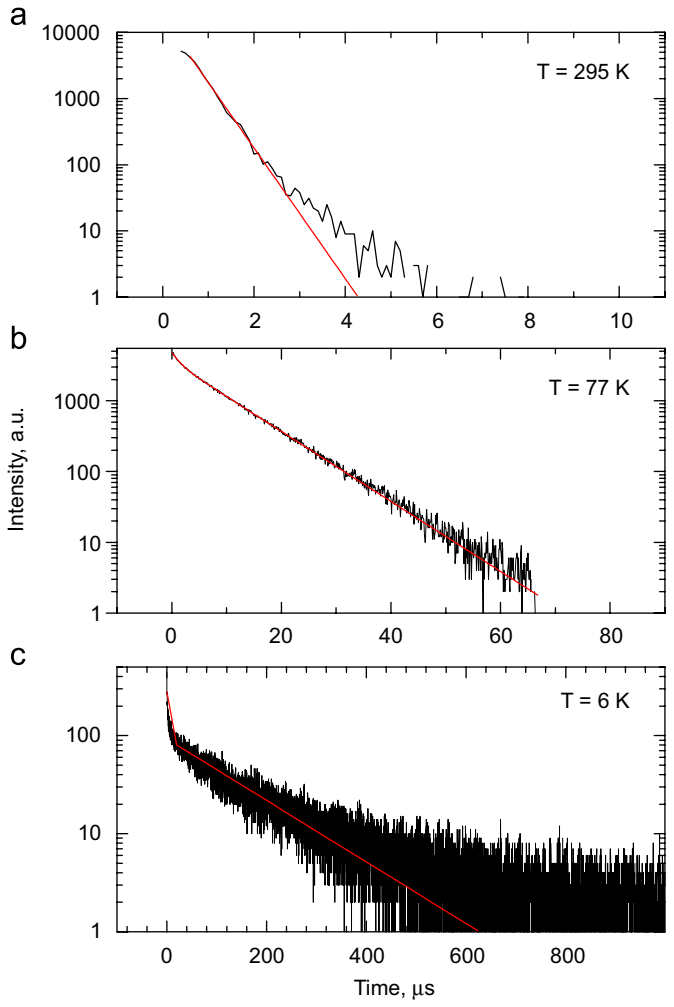


Fig. 2. Scintillation decay curves of BGO measured at $T = 6$ (a), 77 (b) and 295 K (c) using α excitations. The solid line shows the best fit to the experimental data using one (a) and two exponentials (b, c). The parameters of the fit are listed in Table 1.

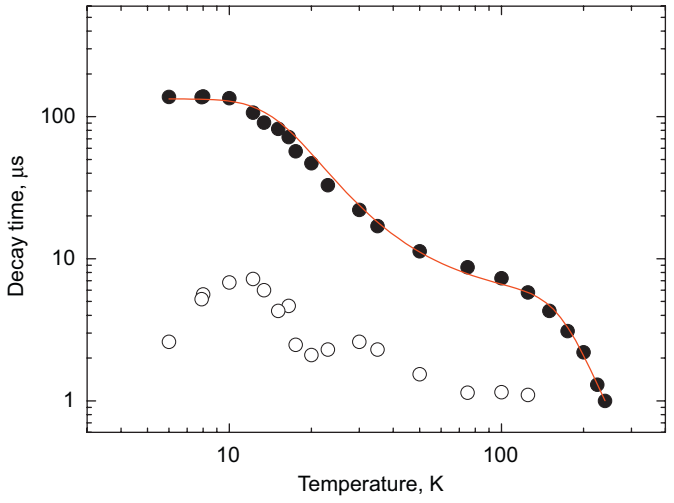


Fig. 3. Temperature dependence of the scintillation decay time constants of BGO (excitation with ^{241}Am α-particles). Open and solid symbol represent the fast and the slow time constant, respectively, obtained from a fit with two exponential functions. The solid curve shows the best fit to the experimental data using the three-level model of emitting centre. Parameters of the fit are given in text.

Table 1
Parameters of scintillation decay curves of BGO at different temperatures obtained using a sum of two exponential functions: $f(t) = (A_1/(A_1 + A_2))e^{-t/\tau_1} + (A_2/(A_1 + A_2))e^{-t/\tau_2}$

Temperature (K)	A_1 (%)	τ_1 (μs)	A_2 (%)	τ_2 (μs)
295	1	0.43 (0.08)	–	–
77	0.24	1.4 (0.04)	0.74	8.7 (0.1)
6	0.68	2.4 (0.04)	0.82	138 (0.6)

two models for the luminescence centre were proposed. The first model argues that the radiative transitions from the upper $^3P_1$ to the ground $^1S_0$ state of the Bi^{3+} ion give rise to the main emission band at 2.5 eV [3,25]. This model is also supported by calculations of the BGO electronic structure, indicating that the bismuth states are responsible for the dominant contribution to the top of valence band and the bottom of the conduction band [26]. The second model assigns the intrinsic luminescence of the crystal to the emission of self-trapped excitons [17,24,27]. Finally, some authors consider both, the mechanism of excitons recombination and the emission of selectively excited Bi^{3+} ions [28]. Though very complete and thorough, these investigations were carried out using selective optical excitations, which is the principal difference with respect to the present studies. One should remember that optical quanta used for luminescence excitation are absorbed in a thin (a few nanometres) layer, while ionising α-particles penetrate ~10 μm below the crystal surface.

The character of the temperature dependence of the decay time constant observed for BGO in the present study is typical for the three-level model of the emitting centre constituting a ground and two excited levels of which the lower one is metastable. Given the electron configuration of ground and excited states of Bi^{3+}, it is logical to assign the luminescence of BGO to this ion. In line with this assumption the emission of Bi^{3+} at low temperatures originates from the $^3P_0$ metastable level, while at higher temperature transitions occurs mainly between excited $^3P_1$ and ground $^1S_0$ state. The respective model predicts the existence of the low-temperature plateau in the $\tau = f(T)$ dependence and the gradual decrease of the decay time constant with temperature, caused by establishing a thermal equilibrium between two emitting levels [29,30]. A further generalisation of the model was achieved when the processes of the non-radiative quenching of excited level to the ground state were taken into consideration [15]. Eventually, the shortening of the decay time constant τ with increasing temperature T can be understood as the result of thermally activated re-population of excited states and their non-radiative de-excitation:

$$\frac{1}{\tau} = \frac{k_1 + k_2 \exp(-D/k_B T)}{1 + \exp(-D/k_B T)} + K \exp(-\Delta E/k_B T). \quad (1)$$

Here k_1 and k_2 are the probabilities of radiative decay from the lower $^3P_0$ and upper $^3P_1$ excited states to the ground $^1S_0$ state, D is energy difference between the excited states, k_B is the Boltzmann constant. The decay rate K and the energy barrier ΔE are the characteristic parameters of the non-radiative luminescence quenching process. It was shown recently that this formula explains well all major features of the observed $\tau = f(T)$ dependence across the entire temperature range of interest in a material characterised by three-level luminescence centres [15].

Formula (1) was used to fit the measured temperature dependence of the slow decay component in BGO. The values obtained from the fit are $k_1 = 7.5 \times 10^3\ s^{-1}$, $k_2 = 4.5 \pm 0.3 \times 10^5\ s^{-1}$, $D = 6.4 \pm 0.5$ meV, $K = 1.3 \pm 0.2 \times 10^8\ s^{-1}$ and $\Delta E = 105 \pm 4$ meV and the calculated theoretical curve demonstrates fairly good agreement with the experimental data points (see Fig. 3).

The three-level model of the emission centre in BGO gives a reasonable interpretation of the observed temperature dependence. Certainly, this is a very realistic and consistent model for crystals with intrinsic emission at low temperatures when the energy transfer processes are suppressed. A sensible approximation in the intermediate temperature range is that the energy transfer due to exciton migration can provide an additional channel for radiative recombination. This channel is important when impurity emission is considered [27] and the model might be improved by including this feature. The derivation of the respective relations should be the subject of the forthcoming studies.

4. Conclusion

Motivated by strong interest in the application of BGO in cryogenic experimental search for rare events we measured for the first time the temperature dependence of the scintillation decay time and the light response of BGO over 6–307 K temperature range. It is demonstrated that the light yield of the crystal increases steadily with cooling down to 100 K and remains practically unchanged at lower temperatures. An estimate of the light yield of BGO at low temperature gives $23{,}700 \pm 2600$ ph/MeV, which is substantially higher than the light yield of a $CaWO_4$ scintillator ($15{,}800 \pm 1600$ ph/MeV) [31]. The results of measurements of the temperature dependence of the decay kinetics were analysed in the framework of the three-level model of the Bi^{3+} emission centre. This model is well suited for the interpretation of the low-temperature properties of this intrinsic scintillator. Given that performance tests of BGO as cryogenic phonon-scintillation detector prove to be very successful [10], we consider this material as a very attractive target for cryogenic experimental rare event research.

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